

BRAIN TUMOR DETECTION (END-TO-END)

1. Algorithm Includes

Step-by-Step Procedure:

1. Collect MRI brain scan images.
2. Preprocess the images.
3. Resize images to a fixed size.
4. Normalize pixel values.
5. Split the dataset into training and testing data.
6. Train the CNN model.
7. Evaluate model performance.
8. Upload new MRI images.
9. Predict whether a tumor is present.
10. Display the prediction result.

Logic or Method Used

The project uses Deep Learning and Convolutional Neural Networks (CNNs). CNN automatically extracts important features from MRI images and learns patterns associated with brain tumors. Based on these patterns, the model classifies images into Tumor and Non-Tumor categories.

Model or Technique Applied

Model: Convolutional Neural Network (CNN)

Techniques Used:

- Image Preprocessing
- Feature Extraction
- Deep Learning Classification
- Model Training and Evaluation

Process Flow of the System

MRI Image Input



Image Preprocessing



CNN Feature Extraction



Model Training



Prediction



Tumor / Non-Tumor Result

2. Dataset Includes

Source of Data

Kaggle Brain MRI Images Dataset

Number of Records / Samples

- Total Images: 3,000+ MRI images
- Tumor Images: 1,500+
- Non-Tumor Images: 1,500+

Features or Attributes

Since it is an image dataset:

- MRI Image
- Label (Tumor / Non-Tumor)

Training and Testing Split

- Training Data: 80%
- Testing Data: 20%

Data Format or Type

- Format: JPG / PNG
- Type: MRI Medical Images